

Q4. Range Sum Query

Solved



Stuck somewhere?  
Ask for help from a TA and get it resolved.

Get help from TA.

Problem Description

You are given an integer array **A** of length **N**.  
You are also given a 2D integer array **B** with dimensions **M x 2**, where each row denotes a [L, R] query.  
For each query, you have to find the sum of all elements from L to R indices in A (1 - indexed).  
More formally, find A[L] + A[L + 1] + A[L + 2] +... + A[R - 1] + A[R] for each query.

Problem Constraints

1 <= N, M <= 10<sup>5</sup>  
1 <= A[i] <= 10<sup>9</sup>  
1 <= L <= R <= N

Input Format

The first argument is the integer array A.  
The second argument is the 2D integer array B.

Output Format

Return an integer array of length M where i<sup>th</sup> element is the answer for i<sup>th</sup> query in B.

Example Input

Input 1:

A = [1, 2, 3, 4, 5]  
B = [[1, 4], [2, 3]]

Input 2:

A = [2, 2, 2]  
B = [[1, 1], [2, 3]]

Example Output

Output 1:

[10, 5]

Output 2:

[2, 4]

Example Explanation

Explanation 1:

The sum of all elements of A[1 ... 4] = 1 + 2 + 3 + 4 = 10.  
The sum of all elements of A[2 ... 3] = 2 + 3 = 5.

Explanation 2:

The sum of all elements of A[1 ... 1] = 2 = 2.  
The sum of all elements of A[2 ... 3] = 2 + 2 = 4.

See Expected Output

Java7 (Open-Jdk-1.7.0)

```
12
13     String[] str1 = inputLine1.split(",");
14     String[] str2 = inputLine2.split(",");
15
16     int totalRows = (str2.length / 2);
17     int N = str1.length;
18
19     int[] input = new int[N];
20     int[][] queries = new int[totalRows][2];
21     long[] answer = new long[totalRows];
22
23     for(int i = 0; i < N; i++)
24     {
25         input[i] = Integer.parseInt(str1[i]);
26     }
```

Test Output Code's All Neat!

Compiling your Code...  
> Success!

Running Test Cases...  
> TestCase - Easy Success  
> TestCase - Hard Success

Final Verdict  
> Correct Answer

Test Test With Custom Input Submit